

# IT 335/IT 3004 - Introduction to Natural Language Processing

Reading Material  
Introduction to Machine Learning

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# What is Machine Learning?

**Machine Learning (ML)** is a core branch of Artificial Intelligence (AI) focused on creating systems that learn from data to make predictions or decisions without being explicitly programmed. Unlike traditional programming, which follows the logic: **input + rules = output**, ML uses **input + output examples to learn rules**, which can then be applied to new data.

Arthur Samuel, a pioneer in ML, described it in 1959 as the ability of computers to *"learn from data without being explicitly programmed."* His checkers-playing program exemplified this idea. Later, Tom Mitchell (1997) formalized the definition: *"A computer program is said to learn from experience (E) with respect to some task (T) and performance measure (P), if its performance at task T, as measured by P, improves with experience E."*

Think of ML like teaching a child. Instead of giving them rules to identify animals, you show them many labeled pictures. Over time, they learn to recognize cats, dogs, and birds without needing an explicit rulebook.

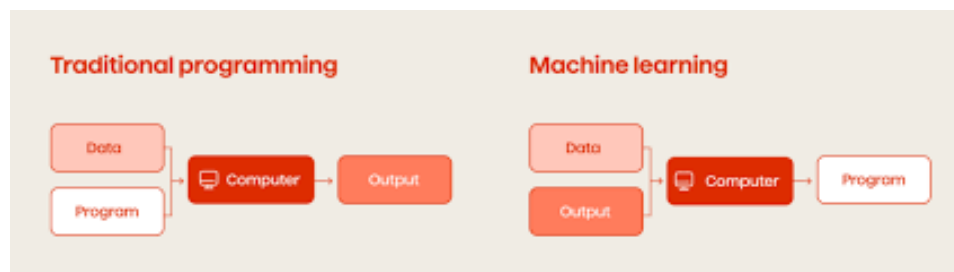


Image 1: A side-by-side diagram of Traditional Programming:  $\text{Input} + \text{Rules} \rightarrow \text{Output}$  and Machine Learning:  $\text{Input} + \text{Output} \rightarrow \text{Learning} \rightarrow \text{Rules}$

# Why Use Machine Learning?

**Machine learning (ML)** is increasingly relied upon because it can solve **complex problems** that traditional rule-based programming struggles with. In many real-world scenarios, defining every rule manually is either infeasible or impossible. For example, human language translation involves intricate grammar rules and countless exceptions. Instead of hard-coding these, machine learning models can analyze large bilingual text datasets to learn translation patterns on their own.

## Key Advantages of Machine Learning

### 1. Adaptability

Traditional systems operate on fixed logic and struggle to adapt when conditions shift. ML systems, on the other hand, continuously learn from new data and improve over time.

- **Example:** Social media platforms like Facebook or TikTok update their content recommendations based on user interactions. Similarly, stock prediction algorithms adjust forecasts as new market data arrives.

## 2. Scalability

Machine learning can analyze vast amounts of data far beyond human capacity.

- **Example:** A human fraud analyst may manually review dozens of suspicious transactions, while an ML model can scan millions in seconds to detect patterns of fraud.

## 3. Handling Unstructured or Ill-Defined Data

Some problems, like image recognition or voice commands, involve unstructured data that's difficult to program rules for.

- **Example:** ML enables voice assistants like Siri or Alexa to interpret spoken language, including varying accents and intonations, and respond appropriately.

## 4. Consistency and Accuracy

ML systems can outperform humans in tasks requiring pattern recognition, especially in noisy or repetitive environments.

- **Example:** In medical imaging, ML models have been shown to detect tumors in X-rays or MRIs with accuracy equal to or better than trained radiologists.

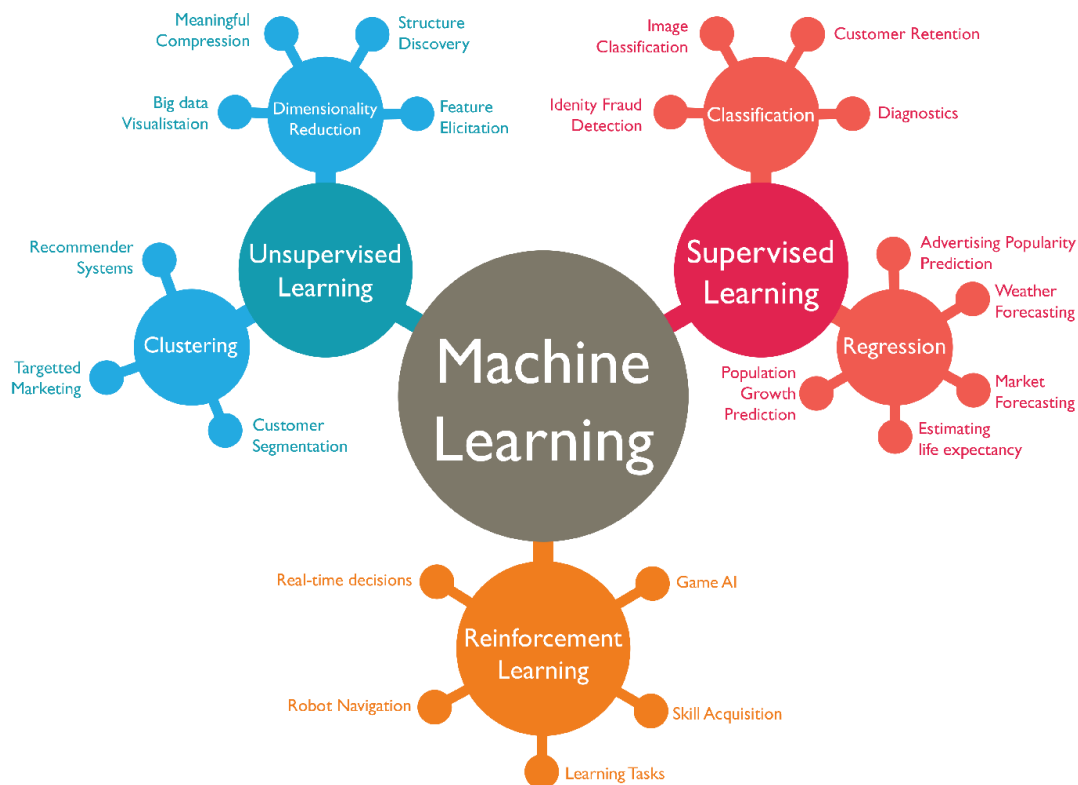
## 5. Personalization

Machine learning tailors services to individual users, enhancing user experience and engagement.

- **Example:** Platforms like Netflix or Spotify use ML algorithms to recommend movies or songs based on viewing/listening history and preferences.

| Use Case           | Description                                 | Example                                                                  |
|--------------------|---------------------------------------------|--------------------------------------------------------------------------|
| Spam Detection     | Identifies and filters unwanted emails.     | Gmail automatically learns and blocks new spam tactics.                  |
| Speech Recognition | Converts spoken language into written text. | Siri and Alexa recognize different accents and dialects.                 |
| Fraud Detection    | Detects unusual financial behavior.         | Banks flag sudden high-value purchases abroad as suspicious.             |
| Healthcare         | Analyzes medical data for diagnosis.        | Detecting tumors in X-rays or MRIs more consistently than human experts. |

|                             |                                                         |                                                                      |
|-----------------------------|---------------------------------------------------------|----------------------------------------------------------------------|
| Recommendation Systems      | Suggests personalized content.                          | Netflix recommends shows based on what you've previously watched.    |
| Autonomous Vehicles         | Makes real-time decisions from sensor input.            | Tesla autopilot detecting pedestrians and other vehicles.            |
| Customer Support Automation | Automates common support queries using chatbots.        | E-commerce platforms deploy AI chatbots for order tracking and FAQs. |
| Predictive Maintenance      | Anticipates equipment failure by analyzing sensor data. | Airlines use ML to predict engine failures before they occur.        |



*Image 2: ML applications*

Machine learning's strength lies in its flexibility, scalability, and ability to learn from data. Whether it's filtering spam, diagnosing disease, driving cars, or recommending a playlist, ML adapts and improves—making it an essential tool in nearly every modern industry.

# Types of Machine Learning Systems

Machine learning systems can be categorized in multiple ways depending on how they learn from data, how they process input, and how they are deployed. Understanding these categories is essential for choosing the right type of algorithm or system for a given task.

## 1. Supervised vs. Unsupervised Learning

### Supervised Learning

Learns from **labeled data** — the system is trained on inputs and their correct outputs. The goal is to learn a mapping function that generalizes to new data.

- **Examples:**
  - Predicting house prices from size, location, etc.
  - Classifying emails as spam or not spam.

### Unsupervised Learning

Learns from **unlabeled data** — the system discovers hidden patterns, structures, or groupings without knowing the correct answer.

- **Examples:**
  - Customer segmentation in marketing.
  - Anomaly detection in cybersecurity.

**Key Difference:** Supervised learning has known outcomes; unsupervised learning does not.

## 2. Semi-Supervised and Self-Supervised Learning

### Semi-Supervised Learning

Uses a small amount of labeled data and a large amount of unlabeled data. Useful when labeling data is expensive.

- **Example:** Face recognition with only some labeled faces.

### Self-Supervised Learning

Generates its own labels from raw input data. Popular in training large language or vision models.

- **Examples:**
  - Predicting the next word in a sentence (used in models like GPT).
  - Identifying whether two images show the same object (contrastive learning).



**Key Idea:** These approaches reduce reliance on costly human labeling.

### 3. Reinforcement Learning (RL)

In reinforcement learning, an agent interacts with an environment and learns by receiving rewards or penalties. The goal is to learn a policy that maximizes long-term reward.

- **Examples:**
  - A robot learning to walk.
  - AlphaGo learning to play Go by competing against itself.

**Key Idea:** Learning through trial-and-error based on feedback from the environment.

### 4. Batch vs. Online Learning

#### Batch Learning

Trains on the entire dataset at once. Good for stable environments where data doesn't change frequently.

- **Example:** Predicting student performance from last year's exam data.

#### Online Learning

Updates the model incrementally with each new data point. Ideal for fast-changing or real-time environments.

- **Examples:**
  - Stock price prediction.
  - Personalized news recommendations.

**Key Difference:** Batch learning is static; online learning is dynamic.

### 5. Instance-Based vs. Model-Based Learning

#### Instance-Based Learning

Stores training examples and makes predictions by comparing new data to stored instances.

- **Example:** k-Nearest Neighbors (k-NN), where predictions are based on the closest data points.

#### Model-Based Learning

Builds a mathematical model from training data and uses it to make generalized predictions.

- **Examples:**

- Linear regression
- Decision trees
- Neural networks

**Key Difference:** Instance-based learning relies on memory; model-based learning relies on abstraction.

## 6. Deterministic vs. Probabilistic Models

### Deterministic Models

Always produce the same output for a given input.

- **Example:** Standard linear regression.

### Probabilistic Models

Produce probability distributions over possible outcomes, capturing uncertainty.

**Examples:**

- Logistic regression
- Bayesian networks

**Key Idea:** Probabilistic models are valuable when uncertainty matters (e.g., medical diagnosis, risk scoring).

## 7. Offline vs. Online Systems (Deployment Perspective)

This distinction refers to how systems behave after training, rather than during training.

### Offline Systems

Trained once and then deployed for use without further updates.

- **Example:** A model trained to predict customer churn and deployed monthly.

### Online Systems (Real-Time)

Continuously update and respond in real time as new data arrives.

**Examples:**

- Fraud detection systems monitoring transactions in real time.
- Self-updating recommendation engines.

**Key Idea:** Offline = static predictions; online = dynamic responses.

# Supervised Learning Examples

Supervised learning is one of the most common types of machine learning. In this approach, the system learns from a dataset where the correct answers (labels) are already provided. There are two major types of supervised learning tasks: classification and regression.

## Classification

Classification is used when the goal is to sort data into categories or classes. Each input is assigned to one of a set of predefined classes.

- **Examples:**
  - A healthcare system classifying patients as diabetic or non-diabetic.
  - An email filter deciding if a message is spam or not.
  - A photo app recognizing whether an image shows a cat, dog, or car.

In classification, the outputs are discrete labels (categories).

## Regression

Regression is used when the goal is to predict a continuous value rather than a category.

- **Examples:**
  - Predicting the selling price of a house based on its size, location, and features.
  - Estimating tomorrow's temperature.
  - Forecasting electricity consumption in a city.

In regression, the output is a number, not a category. In both cases, the model is trained on labeled examples, and then it can make predictions on new, unseen data.

# Unsupervised Learning Examples

Unsupervised learning helps us discover patterns in data without using labeled outcomes. It is especially useful when we don't know in advance what we're looking for. The two main types of tasks in unsupervised learning are clustering and dimensionality reduction.

## Clustering

Clustering involves grouping similar data points together. The system learns which data points are close to each other in terms of features.

- **Examples:**
  - A company analyzing customer buying habits to group customers into segments (e.g., budget buyers, luxury shoppers).



- Social media platforms grouping users by shared interests.
- News websites clustering articles by topic.

Clustering helps find hidden groupings in data.

## Dimensionality Reduction

Some data has many features (for example, each pixel in an image), which can make analysis complex. Dimensionality reduction simplifies the data while keeping its most important features.

- **Examples:**
  - Reducing the number of variables in genetic data to focus on the most important ones.
  - Visualizing large datasets in 2D or 3D using techniques like PCA (Principal Component Analysis).

Dimensionality reduction makes large datasets easier to interpret and process.

## Key Challenges in Machine Learning

Building a good machine learning model isn't always easy. One of the biggest challenges is finding the right balance between a model that's too simple and one that's too complex. This leads to the concepts of underfitting and overfitting.

### Underfitting

Underfitting happens when a model is too simple to capture the patterns in the data. As a result, it performs poorly on both training and new data.

**Example:** Predicting house prices using only the size of the house and ignoring its location and age.

**Solutions:**

- Use a more complex model.
- Add more relevant features to the data.

### Overfitting

Overfitting occurs when a model is too complex and memorizes the training data. It performs well on known data but fails on new data.

**Example:** A spam filter that perfectly classifies training emails but cannot handle new types of spam.

**Solutions:**

- Use simpler models.

- Apply techniques like regularization.
- Train on more data to help the model generalize.

A good ML model must generalize well. It should work not just on the training data, but also on new data.

## Model Evaluation: Testing and Validation

To measure how well a machine learning model performs, we divide our dataset into separate parts:

### Training, Validation, and Test Sets

- **Training Set:** Used to teach the model.
- **Validation Set:** Used to fine-tune model settings (called hyperparameters).
- **Test Set:** Used to evaluate final performance on unseen data.

### Cross-Validation

A common technique is **k-fold cross-validation**:

- The dataset is split into k parts (or "folds").
- The model is trained on k-1 parts and tested on the remaining part.
- This is repeated k times, each time with a different test part.
- The results are averaged for a more reliable estimate of performance.

#### Benefits:

- Reduces the chance that a model performs well just by luck.
- Provides a better understanding of how the model will perform in real-world use.

Proper evaluation ensures that a model works reliably, not just in the lab but in the real world.

## Hyperparameter Tuning and Model Selection

### What Are Hyperparameters?

Hyperparameters are settings chosen before training begins. They are not learned from the data, but they can significantly affect model performance.

- **Examples:**
  - The learning rate in a neural network.
  - The number of clusters in k-Means.
  - The maximum depth of a decision tree.

## Tuning Methods

- **Grid Search:** Tries every possible combination of hyperparameter values.
- **Random Search:** Chooses random combinations to explore.
- **Bayesian Optimization:** Uses probability models to choose the next best combination to try.

## Choosing the Right Model

Different tasks require different models. Choosing the right one depends on:

- Accuracy needs
- Speed of training
- Interpretability
- Available computing resources

**Example:** In healthcare, a simpler model like logistic regression may be preferred over a complex neural network because it's easier to explain to doctors, even if it's slightly less accurate.

Good model choice and hyperparameter tuning can greatly improve machine learning performance.

## Recap: Instance-Based vs. Model-Based Learning

**Instance-based learning methods** store the training data and classify new inputs by comparing them to stored examples. This is similar to human memory: when encountering something new, we compare it to past experiences. **Model-based learning methods**, on the other hand, build a general formula or function that can predict outcomes for new cases. For example, k-Nearest Neighbors recommends restaurants by comparing your choices to those of similar users, while linear regression might predict your salary based on years of experience and education.

## Conclusion

Machine learning is about building systems that learn from data. It helps make decisions, detect patterns, and solve complex problems across many industries. By understanding different types of learning (supervised, unsupervised, reinforcement), common problems (overfitting, underfitting), and important tools (evaluation methods and hyperparameter tuning), students gain a strong foundation to explore advanced topics like deep learning, natural language processing, and AI ethics.